

Color 4 Color

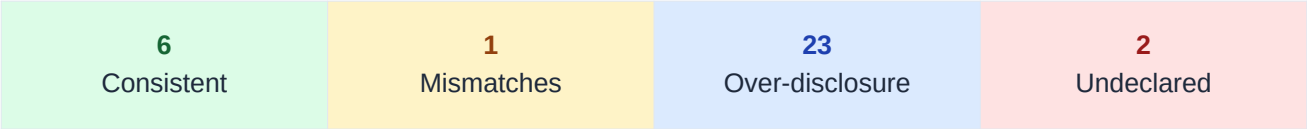
com.Acroit.Color4Color

Developer	Acroit
Genre	GAME_ARCADE
Installs	10+
Audit Date	2026-04-26

Overall Risk Score

0.30

MEDIUM



Research prototype · Ongoing research | Policy: <https://unity.com/legal/game-player-and-app-user-privacy-pol...>

Executive Summary

Color 4 Color declares 6 data type(s) collected and 6 shared in its Play Data Safety label, across 32 data type(s) analyzed. Our heterogeneous GNN audit model identifies **2 potential undeclared collection(s)** — data types implied by embedded SDK signals but absent from both label and policy — and **1 policy-vs-label mismatch(es)**. Overall risk score: **0.30 (MEDIUM)**.

Top discrepancies by risk:

- **installed_apps**: Undeclared Collection (confidence 0.97)
- **crash_logs**: Undeclared Collection (confidence 0.97)
- **diagnostics**: Policy Label Mismatch (confidence 0.99)

Class definitions:

CONSISTENT: Label, policy, and SDK signals agree.

POLICY_LABEL_MISMATCH: Label discloses data not mentioned in policy.

OVER_DISCLOSURE: Policy mentions data not declared in label.

UNDECLARED_COLLECTION: SDK signals collection undisclosed in label and policy.

Discrepancy Table

Sorted by risk class (undeclared collection first). Y/N: label=data-safety label, policy=policy text, sdk=SDK signal.

Data Type	Class	Conf.	Evidence
installed_apps	Undeclared Collection	0.97	label:N policy:N sdk:Y (1, 2, 4 +3)
crash_logs	Undeclared Collection	0.97	label:N policy:N sdk:Y (1, 2, 4 +3)
diagnostics	Policy Label Mismatch	0.99	label:Y policy:N sdk:Y (1, 2, 4 +3)
political_or_religious_beliefs	Over Disclosure	0.97	label:N policy:Y sdk:N (1, 2, 4 +3)
sms_mms	Over Disclosure	0.97	label:N policy:Y sdk:N (1, 2, 4 +3)
date_of_birth	Over Disclosure	0.96	label:N policy:Y sdk:N (1, 2, 4 +3)
biometric_data	Over Disclosure	0.95	label:N policy:Y sdk:N (1, 2, 4 +3)
contacts	Over Disclosure	0.95	label:N policy:Y sdk:N (1, 2, 4 +3)
user_payment_info	Over Disclosure	0.95	label:N policy:Y sdk:N (1, 2, 4 +3)
precise_location	Over Disclosure	0.94	label:N policy:Y sdk:N (1, 2, 4 +3)
address	Over Disclosure	0.93	label:N policy:Y sdk:N (1, 2, 4 +3)
financial_info	Over Disclosure	0.92	label:N policy:Y sdk:N (1, 2, 4 +3)
files_and_docs	Over Disclosure	0.92	label:N policy:Y sdk:N (1, 2, 4 +3)
voice_or_sound_recordings	Over Disclosure	0.92	label:N policy:Y sdk:N (1, 2, 4 +3)
email_address	Over Disclosure	0.92	label:N policy:Y sdk:N (1, 2, 4 +3)
emails	Over Disclosure	0.90	label:N policy:Y sdk:N (1, 2, 4 +3)
sexual_orientation	Over Disclosure	0.90	label:N policy:Y sdk:N (1, 2, 4 +3)
web_browsing_history	Over Disclosure	0.89	label:N policy:Y sdk:N (1, 2, 4 +3)
phone_number	Over Disclosure	0.89	label:N policy:Y sdk:N (1, 2, 4 +3)
videos	Over Disclosure	0.88	label:N policy:Y sdk:N (1, 2, 4 +3)
name	Over Disclosure	0.88	label:N policy:Y sdk:N (1, 2, 4 +3)
health_info	Over Disclosure	0.86	label:N policy:Y sdk:N (1, 2, 4 +3)

Data Type	Class	Conf.	Evidence
photos	Over Disclosure	0.84	label:N policy:Y sdk:N (1, 2, 4 +3)
race_ethnicity	Over Disclosure	0.84	label:N policy:Y sdk:N (1, 2, 4 +3)
fitness_info	Over Disclosure	0.79	label:N policy:Y sdk:N (1, 2, 4 +3)
other_info	Over Disclosure	0.77	label:N policy:Y sdk:N (1, 2, 4 +3)
device_id	Consistent	1.00	label:Y policy:Y sdk:Y (1, 2, 4 +3)
user_ids	Consistent	1.00	label:Y policy:Y sdk:N (1, 2, 4 +3)
app_interactions	Consistent	1.00	label:Y policy:Y sdk:N (1, 2, 4 +3)
advertising_id	Consistent	0.99	label:N policy:Y sdk:Y (1, 2, 4 +3)
approximate_location	Consistent	0.99	label:Y policy:Y sdk:Y (1, 2, 4 +3)
purchase_history	Consistent	0.99	label:Y policy:Y sdk:N (1, 2, 4 +3)

Evidence Trails

Detailed evidence for UNDECLARED_COLLECTION or confidence > 0.85 non-CONSISTENT findings.

Discrepancy: installed_apps — Undeclared Collection (confidence 0.97)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **Yes** — 1, 2, 4, 5, 7 (+1 more)

Recommended action: Add 'installed_apps' to the Play Data Safety label under 'Data Collected' and update the privacy policy to disclose this collection.

Discrepancy: crash_logs — Undeclared Collection (confidence 0.97)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **Yes** — 1, 2, 4, 5, 7 (+1 more)

Recommended action: Add 'crash_logs' to the Play Data Safety label under 'Data Collected' and update the privacy policy to disclose this collection.

Discrepancy: diagnostics — Policy Label Mismatch (confidence 0.99)

Label declares collection: **Yes**

Label declares sharing: **Yes**

Policy text mentions: **No**

SDK collection signals: **Yes** — 1, 2, 4, 5, 7 (+1 more)

Recommended action: Update the privacy policy to disclose 'diagnostics' collection, aligning it with the Play Data Safety label.

Discrepancy: political_or_religious_beliefs — Over Disclosure (confidence 0.97)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **Yes**

SDK collection signals: **No** — 1, 2, 4, 5, 7 (+1 more)

Recommended action: Review whether 'political_or_religious_beliefs' is genuinely collected. If yes, add to label; if no, remove from policy.

Discrepancy: sms_mms — Over Disclosure (confidence 0.97)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **Yes**

SDK collection signals: **No** — 1, 2, 4, 5, 7 (+1 more)

Recommended action: Review whether 'sms_mms' is genuinely collected. If yes, add to label; if no, remove from policy.

Discrepancy: date_of_birth — Over Disclosure (confidence 0.96)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **Yes**

SDK collection signals: **No** — 1, 2, 4, 5, 7 (+1 more)

Recommended action: Review whether 'date_of_birth' is genuinely collected. If yes, add to label; if no, remove from policy.

Discrepancy: biometric_data — Over Disclosure (confidence 0.95)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **Yes**

SDK collection signals: **No** — 1, 2, 4, 5, 7 (+1 more)

Recommended action: Review whether 'biometric_data' is genuinely collected. If yes, add to label; if no, remove from policy.

Discrepancy: contacts — Over Disclosure (confidence 0.95)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **Yes**

SDK collection signals: **No** — 1, 2, 4, 5, 7 (+1 more)

Recommended action: Review whether 'contacts' is genuinely collected. If yes, add to label; if no, remove from policy.

Discrepancy: user_payment_info — Over Disclosure (confidence 0.95)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **Yes**

SDK collection signals: **No** — 1, 2, 4, 5, 7 (+1 more)

Recommended action: Review whether 'user_payment_info' is genuinely collected. If yes, add to label; if no, remove from policy.

Discrepancy: `precise_location` — **Over Disclosure** (confidence 0.94)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **Yes**

SDK collection signals: **No** — 1, 2, 4, 5, 7 (+1 more)

Recommended action: Review whether '`precise_location`' is genuinely collected. If yes, add to label; if no, remove from policy.

Discrepancy: `address` — **Over Disclosure** (confidence 0.93)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **Yes**

SDK collection signals: **No** — 1, 2, 4, 5, 7 (+1 more)

Recommended action: Review whether '`address`' is genuinely collected. If yes, add to label; if no, remove from policy.

Discrepancy: `financial_info` — **Over Disclosure** (confidence 0.92)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **Yes**

SDK collection signals: **No** — 1, 2, 4, 5, 7 (+1 more)

Recommended action: Review whether '`financial_info`' is genuinely collected. If yes, add to label; if no, remove from policy.

Discrepancy: `files_and_docs` — **Over Disclosure** (confidence 0.92)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **Yes**

SDK collection signals: **No** — 1, 2, 4, 5, 7 (+1 more)

Recommended action: Review whether '`files_and_docs`' is genuinely collected. If yes, add to label; if no, remove from policy.

Discrepancy: `voice_or_sound_recordings` — **Over Disclosure** (confidence 0.92)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **Yes**

SDK collection signals: **No** — 1, 2, 4, 5, 7 (+1 more)

Recommended action: Review whether '`voice_or_sound_recordings`' is genuinely collected. If yes, add to label; if no, remove from policy.

Discrepancy: email_address — Over Disclosure (confidence 0.92)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **Yes**

SDK collection signals: **No** — 1, 2, 4, 5, 7 (+1 more)

Recommended action: Review whether 'email_address' is genuinely collected. If yes, add to label; if no, remove from policy.

Discrepancy: emails — Over Disclosure (confidence 0.90)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **Yes**

SDK collection signals: **No** — 1, 2, 4, 5, 7 (+1 more)

Recommended action: Review whether 'emails' is genuinely collected. If yes, add to label; if no, remove from policy.

Discrepancy: sexual_orientation — Over Disclosure (confidence 0.90)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **Yes**

SDK collection signals: **No** — 1, 2, 4, 5, 7 (+1 more)

Recommended action: Review whether 'sexual_orientation' is genuinely collected. If yes, add to label; if no, remove from policy.

Discrepancy: web_browsing_history — Over Disclosure (confidence 0.89)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **Yes**

SDK collection signals: **No** — 1, 2, 4, 5, 7 (+1 more)

Recommended action: Review whether 'web_browsing_history' is genuinely collected. If yes, add to label; if no, remove from policy.

Discrepancy: phone_number — Over Disclosure (confidence 0.89)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **Yes**

SDK collection signals: **No** — 1, 2, 4, 5, 7 (+1 more)

Recommended action: Review whether 'phone_number' is genuinely collected. If yes, add to label; if no, remove from policy.

Discrepancy: videos — Over Disclosure (confidence 0.88)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **Yes**

SDK collection signals: **No** — 1, 2, 4, 5, 7 (+1 more)

Recommended action: Review whether 'videos' is genuinely collected. If yes, add to label; if no, remove from policy.

Discrepancy: name — Over Disclosure (confidence 0.88)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **Yes**

SDK collection signals: **No** — 1, 2, 4, 5, 7 (+1 more)

Recommended action: Review whether 'name' is genuinely collected. If yes, add to label; if no, remove from policy.

Discrepancy: health_info — Over Disclosure (confidence 0.86)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **Yes**

SDK collection signals: **No** — 1, 2, 4, 5, 7 (+1 more)

Recommended action: Review whether 'health_info' is genuinely collected. If yes, add to label; if no, remove from policy.

Methodology

PolicyGraphAudit-RT represents each Android app as a tri-partite heterogeneous graph connecting three layers: **Privacy Policy** text (OPP-115 segment classification), **Play Data Safety Labels** (declared data types and purposes), and **Runtime Evidence** (inferred SDK trackers via Exodus Privacy). A 2-layer HeteroConv R-GCN encodes all three layers jointly; an MLP classifier head predicts discrepancy classes for each (App, DataType) pair.

The model was trained under an edge-masking protocol (30% of label-determining edges removed) to prevent circularity. It achieves **macro F1 = 0.956** on 39 held-out apps (UNDECLARED_COLLECTION F1 = 0.974), trained on 252 apps and 3,202 labeled pairs. Model card: https://github.com/PolicyGraphAudit-RT/reports/blob/main/reports/m5_model_card.md.

Known limitations: (1) Privacy labels are from a 2022 Play Store snapshot. (2) SDK presence is inferred via category-level priors, not measured from APK bytecode. (3) Policy classifier uses MiniLM + logistic regression (OPP-115 macro F1 = 0.70). (4) Labels are rule-derived weak supervision, not human-annotated. (5) English-language Android apps only.

Disclaimer

This report is produced by a research prototype and is provided for informational and research purposes only. It does not constitute legal advice or a regulatory determination. Findings are based on automated weak-supervision labels from publicly available data and have not been reviewed by a privacy legal professional. Research prototype · Ongoing research.